

L03a_Mary_Ann_ITAI1378 – Journal/Summary

6263-ITAI-1378 – Computer Vision

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A01: Colab and Tools Practice Run – Journal & Reflection

What I Did

For this lab, I completed a full walkthrough of setting up GitHub, exploring its interface, creating a repository, and working with my first Jupyter Notebook using Google Colab. I began by creating a GitHub account and familiarizing myself with the dashboard, including repositories, commits, issues, and pull requests. After exploring the interface, I created a new repository named Jupyter-exploration and initialized it with a README file. I edited the README to include a short description of the lab session and committed the changes directly to the main branch, completing my first GitHub commit.

Next, I moved into the Jupyter Notebook portion of the assignment. I chose to use Google Colab, since it provides a built-in Jupyter environment without requiring any installation. I opened Colab, created a new notebook, and explored the interface. I added a Markdown cell and wrote a short message, then added a Code cell and ran a simple Python command using `print("Hello, World!")`. Running the cells helped me understand how Colab executes code and displays output directly beneath each cell.

After completing the notebook, I saved a copy to my Google Drive and then downloaded the .ipynb file to my computer. To upload the notebook to GitHub, I installed GitHub Desktop, signed in, and cloned my Jupyter-exploration repository to my computer. I copied my notebook file into the local repository folder, returned to GitHub Desktop, and saw the notebook listed under “Changes.” I wrote a commit message, committed the file, and pushed it to GitHub. This completed the full workflow of creating, editing, saving, and uploading a Jupyter Notebook to a GitHub repository.

What I Learned

This lab introduced me to several tools that are foundational in modern computing and AI workflows. First, I learned how GitHub works as a version-control platform. I now understand the difference between a repository, a commit, and a push, and how GitHub Desktop simplifies the process of managing files without needing to use command-line tools. I also learned how cloning a repository creates a local copy on my computer, and how committing changes creates a snapshot of my work that can be pushed back to GitHub.

On the Jupyter Notebook side, I learned how notebooks combine code and documentation in a single interactive environment. Markdown cells allow me to write explanations, notes, or instructions, while

code cells allow me to run Python code and immediately see the results. This makes Jupyter an ideal tool for data science, machine learning, and computer vision work. I also learned how to run cells, how to save notebooks, and how to download them for upload into GitHub.

One of the biggest takeaways from this lab was understanding how these tools fit together. GitHub provides version control and storage, while Jupyter/Colab provides the interactive computing environment. Together, they create a workflow that is used by professionals in AI, data science, and software development. I also learned that even though GitHub Desktop is optional, it makes the process of uploading files much easier for beginners.

I did encounter a few small challenges, such as making sure I was uploading the notebook to the correct repository and understanding how cloning works. However, once I walked through the steps slowly, everything made sense. The combination of GitHub Desktop and Colab made the entire process manageable, even for someone new to version control.

Questions or Comments

Overall, this lab was a helpful introduction to tools that I will be using throughout the course. I feel more confident navigating GitHub and working with Jupyter Notebooks. My only question is whether future assignments will require more advanced Git features, such as branching or pull requests, or if GitHub Desktop will continue to be sufficient for the level of work we are doing. I also appreciate that the lab allowed flexibility in choosing between local installation and online environments, since Colab made the process much easier.